Critical Analysis and Reflection of Concept Representation in Language Models

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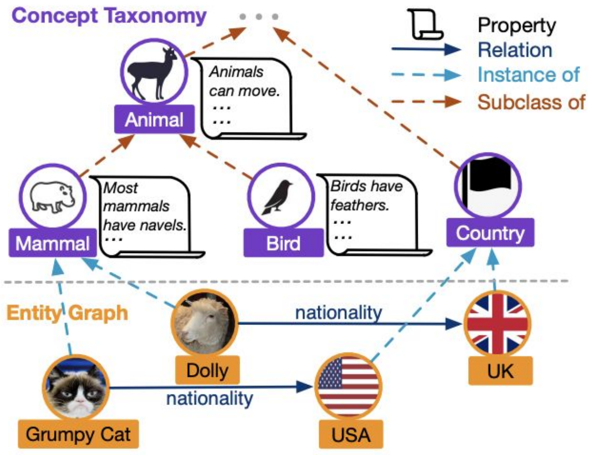
What's Concept Representation in LLM?

What's Concepts?

Concepts are understood as the **mental representations of categories** (e.g., dog, chair, apple) that allow us to *organize knowledge, recognize novel objects, and make predictions* .

"Concepts are the glue that holds our mental world together."[1].



[1] Gregory Murphy. 2002. The Big Book of Concepts. (01 2002).

So the *Problem*

LLMs is brilliant right? **But…**

Their internal concept representations are still ***unclear***.

→

So researchers worked on different ways to find how it's

Why it matters?

Human

Robust and coherent concept organization

Stable across methods, cultures, and languages

Large Language Model

Lack fine-grained **nuance**

Show task-sensitive fragility

Prioritize compression over contextual richness

COPEN Benchmark

Inspired by conceptual schemas in knowledge representation, the authors comprehensively evaluate the conceptual knowledge of pretrained language models (PLMs) by addressing three questions.

(1) Do PLMs organize entities by conceptual similarities? — Conceptual Similarity Judgment (CSJ)
(2) Do PLMs know the properties of concepts? — Conceptual Property Judgment (CPJ)
(3) Can PLMs correctly conceptualize entities in context? — Conceptualization in Contexts (CiC)

Aspect	**Conceptual Similarity Judgment (CSJ)**	**Conceptual Property Judgment (CPJ)**	**Conceptualization in Contexts (CiC)**
Task Type	Multiple-choice classification	Binary sentence classification	Multiple-choice classification
Input	Query entity + candidate entities	Property statement (e.g., “have feathers”)	Sentence containing an entity mention
Output	Most conceptually similar entity	True / False judgment	Most appropriate concept from a concept chain

COPEN Benchmark

Aspect	Conceptual Similarity Judgment (CSJ)	Conceptual Property Judgment (CPJ)	Conceptualization in Contexts (CiC)
Data Collection	Automatic collection + co-occurrence filtering	Automatic collection + human annotation	Sentence collection + human annotation
Key findings	PLMs better distinguish coarse grained concepts	Conceptual transitivity is difficult; conceptual hallucination occurs; word co-occurrence causes hallucination	PLMs over-rely on memory; hierarchy understanding is harder than disambiguation

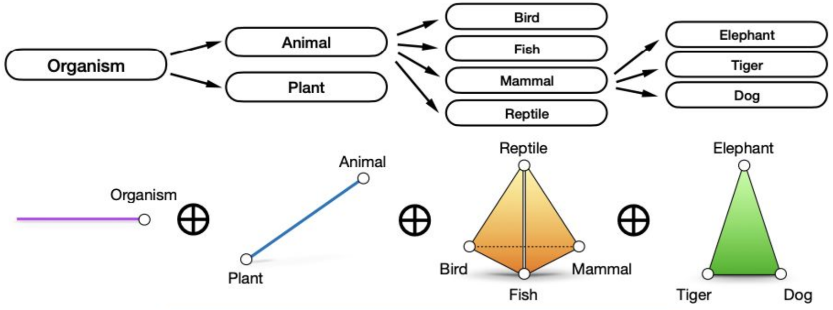
Geometric Analysis of Representations

Treat concepts as geometric objects in embedding space: directions/vectors, subspaces, or polytopes.

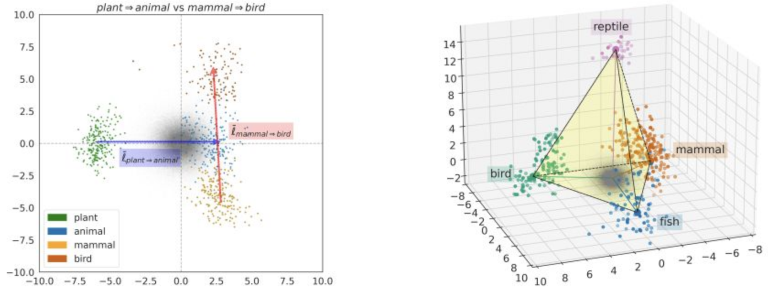
- Binary concepts as *directions to representations* as **vectors**
- Categories as **polytopes**
- Hierarchically related concepts as **orthogonal subspaces**

Estimate concept vectors from WordNet-derived concepts; analyze angles, orthogonality, subspace structure

Gemma, LLaMA-3 models used



(a) Pictorial depiction of the representation of hierarchically related concepts.



(b) Hierarchy is encoded as orthogonality in Gemma. (c) Categorical concepts are represented as polytopes in Gemma.

Evaluation of Semantic Relation Knowledge of PLMs and Humans

Comprehensive evaluation framework to measure how much semantic relation knowledge pretrained language models (PLMs) acquire, and how this compares to humans.

Evaluates 6 relations - hypernymy, hyponymy, holonymy, meronymy, antonymy, synonymy

6 metrics - soundness, completeness, symmetry, asymmetry, **prototypicality, distinguishability**

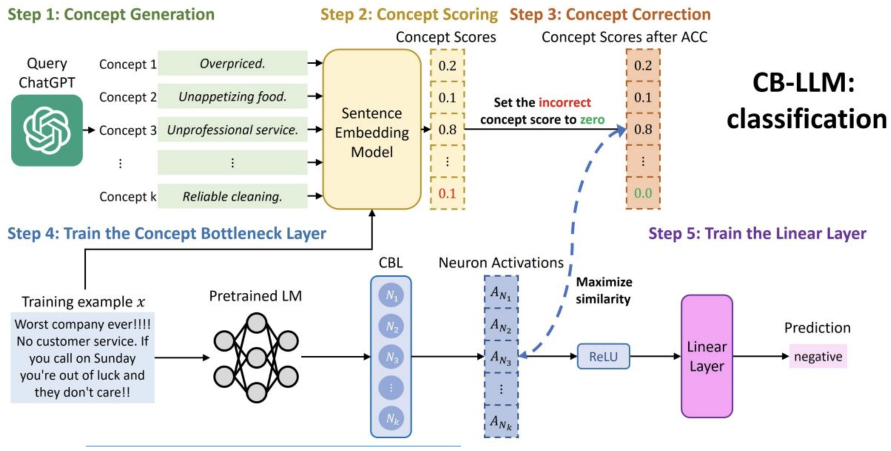
Sample Prompts

[det] [w] is a type of [det] [v]
my favorite [w] is [det] [v]
[det] [w] is a component of [det] [v]
constituents of [det] [w] include [det] [v]
it is not likely to be both [det] [w] and [det] [v]
det] [w] is also known as [det] [v]

Concept Bottleneck Large Language Models

Modify the model architecture to make concepts more explicit and easier to control.

(1) The input text is provided to the model.
(2) The model predicts human-interpretable concepts from the input.
(3) These predicted concepts form a concept bottleneck, forcing the downstream decision to depend explicitly on concept-level representations.
(4) A task classifier operates on the concept representations to produce the final classification output.
(5) The final prediction is interpretable because it can be explained through the intermediate concept predictions.



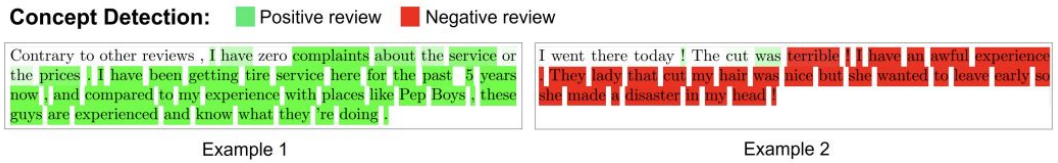


Figure 4: An example of how neurons in CB-LLMs (generation) detect the concepts. A deeper color means higher neuron activations.

Potemkin Understanding in Large Language Models

Benchmarks assume:
If a model can define a concept → it understands it.

Potemkin shows:
A model can explain a concept but fail to apply it correctly.

The paper formalizes this gap using:

Literary techniques
Game theory
Psychological biases

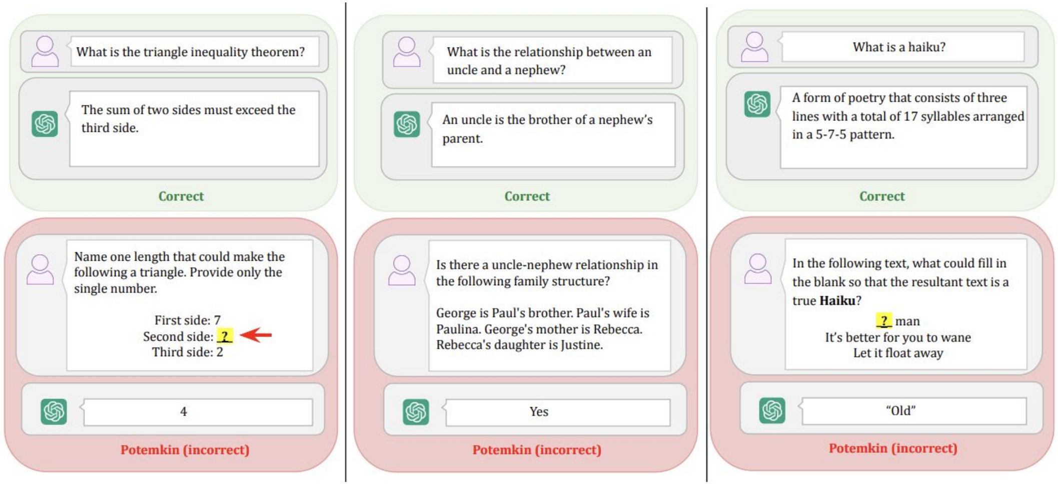


Figure 3. Examples of potemkins. In each example, GPT-4o correctly explains a concept but fails to correctly use it.

Critical Analysis of Methodologies

Comparison of Five Methodological Approaches

Method	Method type	Probing setup	What "concept" means in this method	What the method tests / measures
COPEN Benchmark (Peng et al., 2022)	Task-based probing benchmark	Three tasks (CSJ/CPJ/CiC) evaluated via zero-shot probing, linear probing, and fine tuning	Concept knowledge as behavioral competence across similarity, conceptual properties (incl. transitivity), and context-based conceptualization	Task accuracy across three concept abilities; comparison of probing regimes (zero-shot vs linear vs fine-tune)
The Geometry of Categorical and Hierarchical Concepts in Large Language Models (Park et al., 2024)	Geometric representation analysis	Estimate concept vectors from WordNet-derived concepts; analyze angles, orthogonality, subspace structure	Concepts as vectors; categories as polytopes; hierarchy encoded via structured geometric relations	Alignment between learned concept geometry and WordNet semantic hierarchy (e.g., orthogonality/subspace relations)

Comparison of Five Methodological Approaches

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Relation Knowledge of Pretrained Language Models and Humans (Cao et al., 2024)	Prompt-based benchmarking + human ceiling	Prompt humans and models to produce relata distributions; convert to ranked lists; compute multi-metric scores	Relation knowledge as producing correct relata for a relation and avoiding confusion across relations (meta-relation behavior)	Metrics over ranked lists: soundness, completeness, symmetry/asymmetry + prototypicality and distinguishability
Concept Bottleneck Large Language Models (Sun et al., 2024)	Architectural intervention (concept bottleneck)	Introduce an explicit concept bottleneck before prediction/generation; evaluate performance and interpretability/faithfulness	Concepts as explicit intermediate variables that the model must use to reach outputs	Whether forcing a concept bottleneck yields interpretable decisions while matching black-box performance

Comparison of Five Methodological Approaches

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Potemkin Understanding in LLM (Mancoridis et al., 2025)	Benchmark-validity + consistency diagnostics	Test for explain–apply inconsistencies; use a targeted dataset (three domains) plus a general lower-bound procedure	Understanding as human-like coherence across tasks; “concept” implies consistent explanation and application patterns	Prevalence of “potemkins”: benchmark success paired with non-human failure patterns (especially explain vs apply gaps)

Models and Advantages vs. Disadvantages

Method	Models used (in experiments)	Advantages	Disadvantages
COPEN Benchmark (Peng et al., 2022)	Masked LMs: BERT-base and RoBERTa-base, Autoregressive LMs: GPT-2 and GPT-Neo, Seq2Seq LMs: BART-base and T5	Decomposes concept knowledge into three targeted abilities Compares multiple probing regimes (zero-shot/linear/fine-tune), enabling more robust behavioral interpretation	Still correlational: good performance may come from shortcuts Benchmark scope limitations (language coverage and very-large model access)
The Geometry of Categorical and Hierarchical Concepts in Large Language Models (Park et al., 2024)	Gemma LLaMA-3	Provides a mechanistic, representation level account (vectors/polytopes/orthogonali ty) that can be validated across many concepts; yields interpretable structural predictions	Relies on linear/geometric assumptions; focuses on static representations and WordNetstyle taxonomies, which may not capture context-sensitive concept usage

Models and Advantages vs. Disadvantages

Method	Models used (in experiments)	Advantages	Disadvantages
Evaluation of Semantic Relation Knowledge of Pretrained Language Models and Humans (Cao et al., 2024)	Two model families by objective (MLM vs CLM), - BERT, ALBERT, RoBERTa, OPT variations and also human participants	Adds human ceiling comparison and evaluates beyond hypernymy. Introduces prototypicality and distinguishability metrics to reveal relation confusions and graded structure	Prompt/metric choices and cutoffs can influence outcomes; remains correlational (evaluates outputs rather than causal use of relation knowledge)
Concept Bottleneck Large Language Models (Sun et al., 2024)	Llama3-8B, GPT2, RoBERTa-base, bartlarge-mnli	Causal interpretability leverage: bottleneck forces conceptbased reasoning Matches black-box performance while improving faithfulness rating (classification) and enabling controllable generation	Changes architecture/system design, so conclusions do not directly describe how standard LLMs represent concepts without intervention

Models and Advantages vs. Disadvantages

Method	Models used (in experiments)	Advantages	Disadvantages
Potemkin Understanding in LLM (Mancoridis et al., 2025)	Llama-3.3, Claude-3.5, GPT-4o, GPT-o1-mini, GPT-o3-mini, Gemini-2.0, DeepSeek-V3, DeepSeekR1, Qwen2-VL	Clarifies when human benchmarks fail to test LLM understanding; provides concrete procedures to estimate prevalence of non-human misunderstanding patterns	Targeted procedure captures a specific failure class; general procedure is conservative (lower bound) and may miss many other forms of potemkins

Overview

Probing, relation benchmarks, and geometric analyses are primarily diagnostic (correlational), while CB-LLMs introduce architectural intervention.

Findings depend on assumptions such as linear geometry (geometric method) and prompt/metric design (relation benchmarks).

These 5 methods define concept in different ways and they used different ways to measure that So, we can’t directly compare the results as per my understanding.

Thank you!